Exercise sheet: Week 5 - solutions

1. $X \sim \mathrm{N}(\mu, \sigma^2)$

   $H_0 : \mu = 0, \sigma = 1$ v. $H_1 : \mu = 1, \sigma = 2$.

$$
\begin{aligned}
f(\underline{x}; \mu, \sigma^2) =& (2\pi)^{-\frac{n}{2}} \sigma^{-n} \exp\left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right\} \\
\therefore \frac{L(\mu_0, \sigma_0^2)}{L(\mu_1, \sigma_1^2)} =& \left(\frac{\sigma_0}{\sigma_1} \right)^{-n} \exp\left\{ -\frac{1}{2\sigma_0^2} \sum (x_i - \mu_0)^2 + \frac{1}{2\sigma_1^2} \sum (x_i - \mu_1)^2 \right\} \\
=& \left(\frac{1}{2} \right)^{-n} \exp\left\{ -\frac{1}{2} \sum x_i^2 + \frac{1}{8} \sum (x_i - 1)^2 \right\} \\
=& \left(\frac{1}{2} \right)^{-n} \exp\left\{ -\frac{3}{8} \sum x_i^2 - \frac{1}{4} \sum x_i + \frac{1}{8} n \right\}
\end{aligned}
$$

   Critical region of the MP test is

$$
\begin{aligned}
R =& \left\{ \underline{x} : \frac{L(\mu_0, \sigma_0^2)}{L(\mu_1, \sigma_1^2)} \leq k_1 \right\} \\
=& \left\{ \underline{x} : \exp\left\{ -\frac{3 \sum x_i^2 + 2 \sum x_i}{8} \right\} \leq k_2 \right\} \\
=& \underline{\underline{\left\{ \underline{x} : 3 \sum x_i^2 + 2 \sum x_i \geq k_3 \right\}}}
\end{aligned}
$$

2. a) μ in Poi(μ): $f(x; \mu) = \dfrac{\mu^x e^{-\mu}}{x!}$

$$
\therefore f(x; \mu) = e^{-\mu} \frac{1}{x!} \exp\{(\log \mu) x\}
$$

   i.e. from the exponential family. Also $\log \mu$ is monotonic increasing. Therefore Poisson family has the MLR property through $\sum x_i$. Therefore UMP test of one-sided hypotheses <u>exist</u>.

   b) p in Bin(n, p)

$$
\begin{aligned}
f(x; p) =& \binom{n}{x} p^x (1-p)^{n-x} \\
=& (1-p)^n \binom{n}{x} \left(\frac{p}{1-p} \right)^x \\
=& (1-p)^n \binom{n}{x} \exp\left\{ \log\left(\frac{p}{1-p} \right) x \right\}
\end{aligned}
$$

i.e. from the exponential family.

Q: Is $\log\left(\frac{p}{1-p}\right)$ monotonic for $0 < p < 1$?

$$p \to 0 : \log\left(\frac{p}{1-p}\right) \to -\infty, p \to 1 : \log\left(\frac{p}{1-p}\right) \to +\infty$$

$$\frac{d}{dp}\log\left(\frac{p}{1-p}\right) = \frac{1-p}{p} \cdot \frac{(1-p)\cdot 1 - p(-1)}{(1-p)^2} = \frac{1}{p(1-p)}$$

No turning points, hence it is monotonic increasing (from $-\infty$ to $+\infty$). Therefore Binomial has the MLR property through $\sum x_i$. Therefore UMP test of one-sided hypotheses exist.

c) p in Geo(p)

$$\begin{aligned} f(x;p) =& p(1-p)^{x-1} \\ =& \frac{p}{1-p} \times 1 \times \exp\{\log(1-p)x\} \end{aligned}$$

i.e. from the exponential family. Also $\log(1-p)$ is monotonic decreasing. Therefore Geometric has the MLR property through $\sum x_i$. Therefore UMP test of one-sided hypotheses exist.

3. a) From 2)(a), we have shown that the Poisson family has the MLR property through the sufficient statistic (of μ) $\sum x_i$. It can also be shown that

$$\begin{aligned} \frac{L(\mu_0)}{L(\mu_1)} &= \frac{(\mu_0)^{\sum x_i}\exp(-n\mu_0)}{\prod x_i!} \cdot \frac{\prod x_i!}{(\mu_1)^{\sum x_i}\exp(-n\mu_1)} \\ &= \left(\frac{\mu_0}{\mu_1}\right)^{\sum x_i} \exp\{n(\mu_1 - \mu_0)\} \end{aligned}$$

Since $\mu_1 > \mu_0$, it is clear that the likelihood ratio is monotone decreasing in $\sum x_i$. Hence, the critical region simplifies to

$$\left\{\underline{x} : \frac{L(\mu_0)}{L(\mu_1)} \leq k_1\right\} \Rightarrow \left\{\underline{x} : \sum x_i \geq k\right\}$$

b) Since independent Poisson random variables are closed under addition, we know that $\sum X_i \sim \text{Poi}(n\mu_0)$ under the null hypothesis H_0. Therefore, the probability of a Type-I error is

$$\mathbf{P}(\underline{X} \in R | H_0) = \mathbf{P}\left(\sum X_i \geq k | \mu = \mu_0\right)$$

The following R code computes the Type-I error probability for any value of k:

```
type_1_err_prob <- function(k, n = 10, mu0 = 2){
return(ppois(k - 1, n*mu0, lower.tail = FALSE))
}
```

The Type-I error probabilities are computed as (i) 0.07789, (ii) 0.05248, (iii) 0.03433, and (iv) 0.03433.

It is **not possible** to obtain a Type-I error probability equal to exactly 5%, since the cdf of the Poisson distribution is a step function and it does not map 'onto' the set $[0, 1]$.

c) We first identify a partition for the space of $\{\sum X_i\}$, i.e., $\{\sum X_i < k\}$, $\{\sum X_i > k\}$, and $\{\sum X_i = k\}$. Using the partition theorem (law of total probability), the probability of a Type-I error is

$$\begin{aligned}
&\mathbf{P}(\text{Reject } H_0|\mu = \mu_0) \\
&= \mathbf{P}\left(\sum X_i < k|\mu = \mu_0\right)\mathbf{P}\left(\text{Reject } H_0|\sum X_i < k \cap \mu = \mu_0\right) \\
&\quad + \mathbf{P}\left(\sum X_i > k|\mu = \mu_0\right)\mathbf{P}\left(\text{Reject } H_0|\sum X_i > k \cap \mu = \mu_0\right) \\
&\quad + \mathbf{P}\left(\sum X_i = k|\mu = \mu_0\right)\mathbf{P}\left(\text{Reject } H_0|\sum X_i = k \cap \mu = \mu_0\right) \\
&= 0 + \mathbf{P}\left(\sum X_i > k|\mu = \mu_0\right)\cdot 1 + \mathbf{P}\left(\sum X_i = k|\mu = \mu_0\right)\cdot p
\end{aligned}$$

giving the required result. It is also possible to use the law of total expectation

$$\begin{aligned}
&\mathbf{P}(\text{Reject } H_0|\mu = \mu_0) \coloneqq \mathbf{E}\left[d(\underline{X})|\mu = \mu_0\right] \\
&= \mathbf{P}\left(\sum X_i < k|\mu = \mu_0\right)\mathbf{E}\left[d(\underline{X})|\sum X_i < k \cap \mu = \mu_0\right] \\
&\quad + \mathbf{P}\left(\sum X_i > k|\mu = \mu_0\right)\mathbf{E}\left[d(\underline{X})|\sum X_i > k \cap \mu = \mu_0\right] \\
&\quad + \mathbf{P}\left(\sum X_i = k|\mu = \mu_0\right)\mathbf{E}\left[d(\underline{X})|\sum X_i = k \cap \mu = \mu_0\right] \\
&= \mathbf{P}(\sum X_i > k|\mu = \mu_0) + \mathbf{P}(\sum X_i = k|\mu = \mu_0)\cdot p
\end{aligned}$$

An alternative way is to work out the probability using the rejection region

provided, i.e.,

$$\begin{aligned}
&\mathbf{P}(\text{Reject } H_0|\mu=\mu_0)\\
&=\mathbf{P}(\underline{X}\in R|\mu=\mu_0)\\
&=\mathbf{P}\left[\left\{\sum X_i>k\right\}\cup\left(\left\{\sum X_i=k\right\}\cap A\right)\mid\mu=\mu_0\right]\\
&=\mathbf{P}\left(\sum X_i>k\mid\mu=\mu_0\right)+\mathbf{P}\left(\left\{\sum X_i=k\right\}\cap A\mid\mu=\mu_0\right)\\
&=\mathbf{P}\left(\sum X_i>k\mid\mu=\mu_0\right)+\mathbf{P}\left(\sum X_i=k\mid\mu=\mu_0\right)\cdot p
\end{aligned}$$

Note that the last line follows from the chain rule of probability $\mathbf{P}(B\cap A|C)=\mathbf{P}(B|C)\cdot\mathbf{P}(A|B\cap C)$.

d) We want to identify the values of k and p such that $\mathbf{P}(\text{Type-I error})=0.05$, i.e.,

$$0.05=\mathbf{P}\left(\sum X_i>k\mid\mu=2\right)+p\times\mathbf{P}\left(\sum X_i=k\mid\mu=2\right)$$

By the properties of probability measures, it is clear that we must have $\mathbf{P}\left(\sum X_i>k\mid\mu=2\right)<0.05$. We shall proceed with trial and error, i.e., experiment with different values of k until the probability $\mathbf{P}\left(\sum X_i>k\mid\mu=2\right)$ falls just below 0.05.

$$\begin{aligned}
\mathbf{P}\left(\sum X_i>26\mid\mu=2\right)&=0.07789\\
\mathbf{P}\left(\sum X_i>27\mid\mu=2\right)&=0.05248\\
\mathbf{P}\left(\sum X_i>28\mid\mu=2\right)&=0.03433
\end{aligned}$$

It seems that $k=28$ would be a reasonable choice.

The value of p is hence

$$p=\frac{0.05-\mathbf{P}\left(\sum X_i>28\mid\mu=2\right)}{\mathbf{P}\left(\sum X_i=28\mid\mu=2\right)}=0.86330$$

Conclusion: To achieve a test size of exactly 5%, we should adopt a randomised hypothesis test with rejection threshold $k=28$, and a randomisation trial with probability of rejection equal to $p=86.33\%$.

4. $X\sim\mathrm{N}(\mu,\sigma^2)$ $H_0:\mu=\mu_0$ v. $H_1:\mu\neq\mu_0$

$\Omega=\{-\infty<\mu<\infty,\sigma^2>0\}$ and $\omega=\{\mu=\mu_0,\sigma^2>0\}$

$$L(\mu, \sigma^2) = (2\pi)^{-\frac{n}{2}} (\sigma^2)^{-\frac{n}{2}} \exp\left\{ -\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right\}$$

<u>In ω:</u> $\mu = \mu_0$ and we must find ${\hat{\sigma_0}}^2$.

Using $l(\mu, \sigma^2) = \log L(\mu, \sigma^2)$ and solving $\frac{\partial l}{\partial \sigma^2} = 0$ gives
$\hat{\sigma}_0^2 = \frac{1}{n} \sum (x_i - \mu_0)^2$.

$$\therefore L(\hat{\mu}_0, \hat{\sigma}_0^2) = (2\pi)^{-\frac{n}{2}} \left[\frac{1}{n} \sum (x_i - \mu_0)^2 \right]^{-\frac{n}{2}} \exp\left(-\frac{n}{2} \right)$$

<u>In Ω:</u> we must find $\hat{\mu}, \hat{\sigma}^2$. Again solving $\frac{\partial l}{\partial \mu} = 0$ and $\frac{\partial l}{\partial \sigma^2} = 0$ gives $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$.

$$\therefore L(\hat{\mu}, \hat{\sigma}^2) = (2\pi)^{-\frac{n}{2}} \left[\frac{1}{n} \sum (x_i - \bar{x})^2 \right]^{-\frac{n}{2}} \exp\left(-\frac{n}{2} \right)$$

$$\therdot \lambda(\underline{x}) = \left[\frac{\sum (x_i - \mu_0)^2}{\sum (x_i - \bar{x})^2} \right]^{-\frac{n}{2}}$$

Now

$$\begin{aligned} \sum (x_i - \mu_0)^2 &= \sum \{ (x_i - \bar{x}) + (\bar{x} - \mu_0) \}^2 \\ &= \sum (x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2 + 2(\bar{x} - \mu_0) \sum (x_i - \bar{x}) \\ &= \sum (x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2 + 0 \end{aligned}$$

$$\begin{aligned} \therefore \lambda(\underline{x}) &= \left[1 + n \frac{(\bar{x} - \mu_0)^2}{(n-1)s^2} \right]^{-\frac{n}{2}} \quad \text{as} \quad s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 \\ &= \left[1 + \frac{1}{n-1} [t(\underline{x})]^2 \right]^{-\frac{n}{2}} \end{aligned}$$

with $t(\underline{x})$ as specified. Therefore GLRT has a critical region

$$\begin{aligned} R &= \{ \underline{x} : \lambda(\underline{x}) \leq k_1 \} \\ &= \{ \underline{x} : [t(\underline{x})]^2 \geq k_2 \} \\ &= \underline{\underline{\{ \underline{x} : |t(\underline{x})| \geq k_3 \}}} \end{aligned}$$

For size α, k_3 is such that $\mathbb{P}(\underline{x} \in R \mid (\mu, \sigma^2) \in \omega) \leq \alpha$ i.e.

$$\mathbb{P}(t(\underline{x}) \geq k_3 \cup t(\underline{x}) \leq -k_3) \leq \alpha.$$

Since $t(\underline{x}) \sim t_{n-1}$ when $\mu = \mu_0$ is in ω, we have

$$\begin{aligned} \mathbb{P}(t(\underline{x}) \geq k_3) + \mathbb{P}(t(\underline{x}) \leq -k_3) &\leq \alpha \\ \implies 2 \times \mathbb{P}(t(\underline{x}) \geq k_3) &\leq \alpha \\ \implies \mathbb{P}(t(\underline{x}) \geq k_3) &\leq \frac{\alpha}{2} \end{aligned}$$

by symmetry of the t-distribution.

$\therefore k_3 = t_{n-1}(\frac{\alpha}{2})$ – percentage point from t_{n-1} tables.